

The Laboratory Process

Standard, checkpoint register and data specification

LAB-2026A · version 1 · for signature

25	98	4
laboratory checkpoints	parameter readings per batch	checkpoints that hold production

This document describes what the laboratory measures, when, on what sample, to what threshold, and what happens when a reading falls outside it — across one 470-hour batch. It ends with 189 questions about what the laboratory does, and how long each of it takes.

Prepared 30 August 2026 · companion to PROCESS-2026C, the 470-hour standard

0 • How to read this document

This is a specification, not a narrative. Every statement in it is one of three things, and the difference matters more than anything else here.

This document says	Which means
CONFIRMED	The factory or the laboratory has stated it. Build on it.
CONFIGURABLE	A number or a rule that is deliberately stored as data. It can be changed by an administrator without a code release. The value written here is the starting value, not a commitment.
UNRESOLVED	Nobody has decided it yet. It is written down so it cannot be forgotten. Nothing may be built on it, and no gate may depend on it.
ABSENT	The process does not define this at all. Recorded here so that its absence is a decision and not an oversight.

The rule that governs this whole document

Where this document is silent, the laboratory decides — and then this document is updated in the same week.

Where this document is wrong, the laboratory is right. Tell us, and it changes.

What this document must never be is quietly disagreed with. A reading taken to a rule that is not written here is a reading nobody can defend six months from now.

Who this is for

The laboratory	Read sections 1 to 7. Sign section 15. Answer the questions in the appendix — they are the real deliverable.
The operator	Read section 2 and section 13. You need to know which four checkpoints can stop you.
Management	Read sections 0, 2, 12 and 14. Section 14 is the list of things nobody has decided.
The build team	Read sections 6, 8, 9, 10 and 12. Section 12 is why you can freeze the backend today.

What is deliberately not in here

Temperatures, turner passes, bunker hours and every other production number live in the companion document PROCESS-2026C — the 470-hour standard. This document touches production timing only where a laboratory reading gates it. If a number appears in both documents and they disagree, PROCESS-2026C governs and this document is the one with the error.

1 • The shape of the laboratory’s work

Across one batch the laboratory is asked for 25 distinct checkpoints. Together they produce 98 parameter readings before per-pile expansion — and rather more than that after it, because two of the checkpoints are taken once per pile rather than once per batch.

Kind	Count	What it does
GATE	4	Production stops. The next activity cannot start until an approved result exists.
DECISION	1	Production does not stop, but the reading chooses which branch is taken next.
RECORD	20	Captured for the batch file and for the trend. Never blocks anything.

That ratio is the point. Four checkpoints out of twenty-five can hold the factory. Everything else is evidence. A laboratory that can stop production at twenty-five points will be routed around within a month; one that can stop it at four will be respected at all four.

The laboratory checkpoint spine — 25 checkpoints across one 470-hour batch

4 gates hold production · 1 decision chooses the branch · 20 records are captured and never block

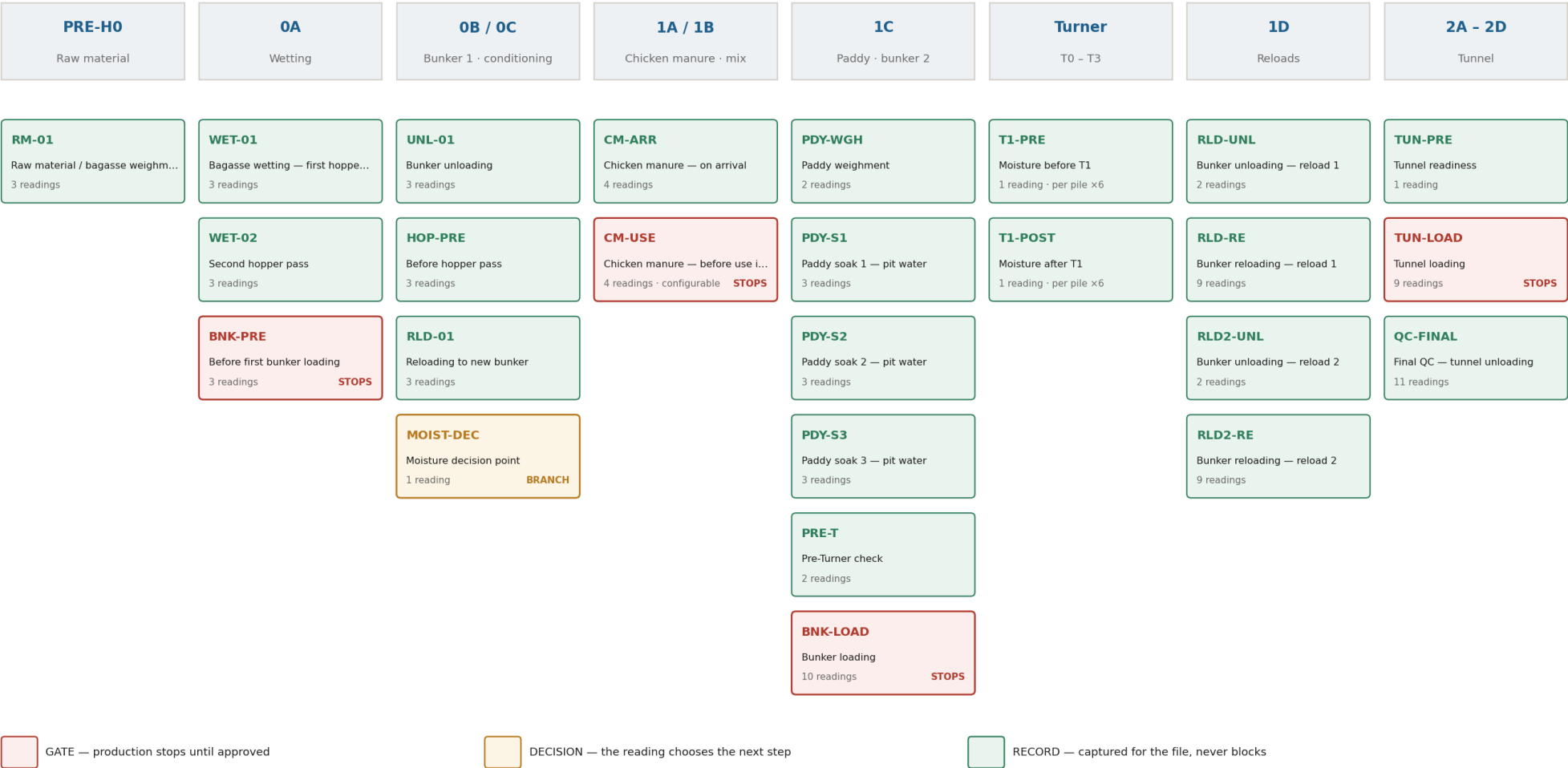


Figure 1 — every laboratory checkpoint in one batch, in process order, coloured by what it can do

2 • The three kinds of checkpoint

This section defines the words. They are used with exactly these meanings everywhere else in the document and everywhere in the system.

GATE

A gate is a hard stop. The named production activity is locked and the operator cannot start it. The lock lifts when an approved laboratory result exists for the gate on that batch — approved, not merely entered.

The four gates	LAB-BNK-PRE — Before first bunker loading
	LAB-CM-USE — Chicken manure — before use in mixing
	LAB-BNK-LOAD — Bunker loading
	LAB-TUN-LOAD — Tunnel loading
What lifts it	An approved result. Entry alone does not lift a gate.
Who can override	UNRESOLVED. Whether a gate can be overridden at all, and by whom, is question M-4 in the appendix. Until it is answered the system implements no override.
What the operator sees	The activity tile is disabled and carries one finished sentence: "Waiting for laboratory approval — moisture, pH, EC." Never a code, never a raw condition.

DECISION

A decision does not stop production. It chooses. The reading is taken, and its value determines which of two or more defined paths the batch takes next. There is exactly one of these in the current process — the moisture decision at LAB-MOIST-DEC, described in section 5.

RECORD

A record is captured because it will be wanted later: to defend a batch, to explain a poor yield, to build the trend that eventually tells the factory what "normal" looks like. It never blocks. A record that was not taken is a hole in the batch file, and the system will show it as a hole — but it will not stop the line.

Why records still matter, said plainly

The factory has no trend today. Not because nobody measured, but because the measurements were never in one shape that could be compared across batches.

Twenty records per batch, taken the same way every time, for twenty batches, is four hundred readings — and that is the first honest picture of the process this factory will ever have.

That is the whole reason for the 20 records. Not compliance. The trend.

3 • The Turner rule

Stated by the laboratory, 30 August 2026

Laboratory testing is performed before T1 and after T1 only.

There is no laboratory test after T0, T2 or T3.

This is written as its own section because it is the rule most likely to be violated by someone acting reasonably. Four turns exist; a diligent person building a screen will produce four test slots, because four turns obviously means four tests. The process says two.

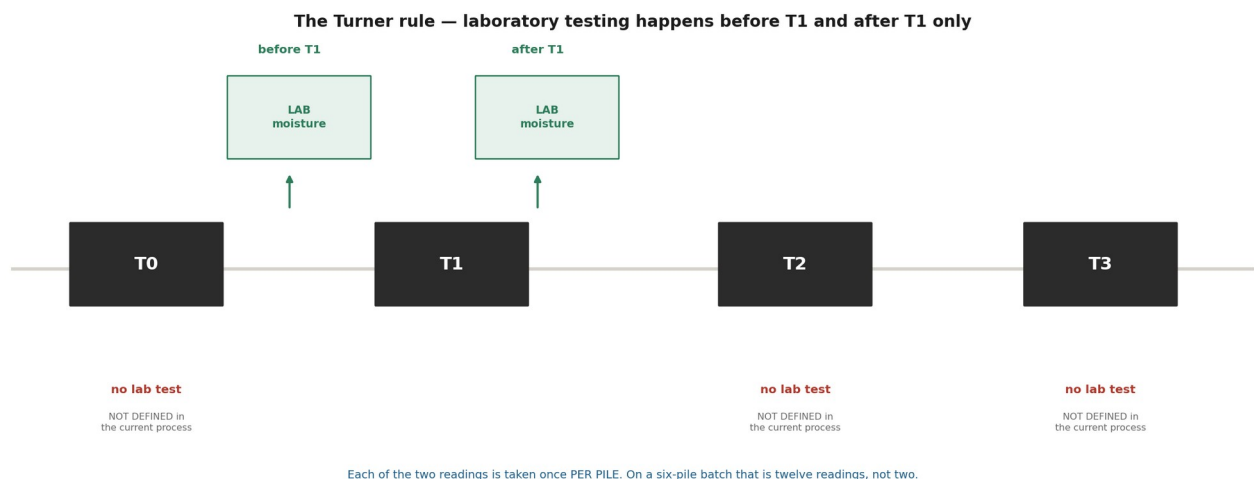


Figure 2 — the two laboratory readings across four Turner passes

What is absent, and recorded as absent

The following are not defined in the current process. They are listed so that their absence is visible and deliberate, and so that adding one later is a change somebody signs rather than a screen somebody drew.

Code	Would be	Status
LAB-T0-POST	Moisture after T0	NOT DEFINED in the current process
LAB-T2-POST	Moisture after T2	NOT DEFINED in the current process
LAB-T3-POST	Moisture after T3	NOT DEFINED in the current process

Per pile, not per batch

Both Turner readings are taken once per pile. On a six-pile batch that is twelve readings, not two. A screen that offers one moisture box at "before T1" is wrong, and the wrongness will not be visible until someone asks why pile 4 has no history.

Checkpoint	Readings	Note
LAB-T1-PRE	6	One per pile. Each tied to its own pile identity, not to the batch.
LAB-T1-POST	6	One per pile, after the T1 pass on that pile.

M1 and M2 are turner machines 1 and 2. A pass is recorded against whichever machine performed it. Which machine turned a pile does not change what the laboratory measures — but it is captured, because the first question after an unexplained variance will be whether the two machines behave the same.

4 • The chicken-manure rule

Chicken manure is tested twice: on arrival, and again before it is used in the mix. The second test is a gate — the mix cannot start without it.

Checkpoint	What is measured
LAB-CM-ARR	Moisture · pH · Nitrogen · Ash. Taken on arrival, on the lot as received. RECORD.
LAB-CM-USE	Moisture · pH · Nitrogen · Ash. Taken on unload for the final mix. GATE.

The exception, and it is configurable

Where the manure arrives less than 12 hours before it is used, LAB-CM-USE is skipped and the arrival reading stands. The material has not had time to change; a second test would measure the same lot twice and delay the mix for nothing.

The 12 is a stored number, not a written-in rule. An administrator can change it to 8, or to 24, without a code release and without a new APK. If the laboratory later decides the window depends on season or on covered versus uncovered storage, that is a change to the rule row, not to the product.

How the system behaves

Gap ≥ window	LAB-CM-USE is required. The mix activity is locked until an approved result exists.
Gap < window	LAB-CM-USE is marked NOT REQUIRED on this batch, with the reason and the measured gap recorded. The mix is not locked.
What is stored	Either way the batch file records which path was taken and why. "Skipped" is a recorded outcome, never a missing row.

Two questions in the appendix (section K) exist only to pin this down: whether the window is measured from arrival to unload or from arrival to mixing start, and whether covered storage changes it.

5 • The moisture decision

One checkpoint in the process is a decision rather than a gate: after the second conditioning cycle, moisture determines whether the batch proceeds or is misted and re-measured.

Reading	Path	What happens
≥ 68 %	PROCEED	The batch moves to the next stage. Nothing is held.
< 67 %	MIST	Controlled misting, then re-measure. The re-measure is a new reading, not an edit of the first.
67 - 68 %	UNRESOLVED	Nobody has decided. The system records the reading, flags it, and refers it. It does not choose.

Why the gap is left open rather than closed with a sensible default

The obvious fix is to write ≥ 67.5 and move on. That fix is invisible six months later, and it will be defended as though the factory chose it.

A number nobody decided, sitting inside a rule everybody trusts, is the most expensive kind of wrong. So the band stays UNRESOLVED, the screen says so in words, and the reading goes to a human until the laboratory rules on it.

Question L-2 in the appendix asks for that ruling. It is one of the four things this project may not decide for itself.

Every re-measure after misting is a new record with its own timestamp and its own photograph. The first reading is never overwritten, never corrected, never deleted. The batch file shows both, in order, and shows the misting between them.

6 • The checkpoint register

All 25 checkpoints, in process order. This table is the contract: it is what the system builds, what the laboratory signs, and what an auditor is shown.

Code	Checkpoint	Stage	Kind	Readings taken	Evidence	Note
LAB-RM-01	Raw material / bagasse weighment	PRE-H0	RECORD	Moisture (%) · pH · Dry weight (kg)	LOT_PHOTO · WEIGHMENT_SLIP	ADVISORY. Recorded, never gates H0.
LAB-WET-01	Bagasse wetting — first hopper pass	0A	RECORD	Moisture (%) · pH · EC (mS/cm)	SAMPLE_PHOTO	Target moisture 69–71 %.
LAB-WET-02	Second hopper pass	0A	RECORD	Moisture (%) · pH · EC (mS/cm)	SAMPLE_PHOTO	
LAB-BNK-PRE	Before first bunker loading	0A→0B	GATE	Moisture (%) · pH · EC (mS/cm)	SAMPLE_PHOTO	Bunker filling stays locked until approved.
LAB-UNL-01	Bunker unloading	0C	RECORD	Moisture (%) · pH · EC (mS/cm)	SAMPLE_PHOTO	
LAB-HOP-PRE	Before hopper pass	0C	RECORD	Moisture (%) · pH · EC (mS/cm)	SAMPLE_PHOTO	
LAB-RLD-01	Reloading to new bunker	0C	RECORD	Moisture (%) · pH · EC (mS/cm)	SAMPLE_PHOTO	
LAB-CM-ARR	Chicken manure — on arrival	1A	RECORD	Moisture (%) · pH · Nitrogen (%) · Ash (%)	LOT_PHOTO	Initial composition of the incoming lot.
LAB-CM-USE	Chicken manure — before use in mixing	1A→1B	GATE	Moisture (%) · pH · Nitrogen (%) · Ash (%)	SAMPLE_PHOTO	CONFIGURABLE. Required only when the gap since arrival exceeds the configured window. Where arrival is under 12 h before use, this checkpoint is skipped and the arrival reading stands.
LAB-MOIST-DEC	Moisture decision point	0C→1B	DECISION	Moisture (%)	SAMPLE_PHOTO	≥ 68 % proceed. < 67 % controlled mist, then re-measure. 67–68 % is UNRESOLVED — record, flag, refer.
LAB-PDY-WGH	Paddy weighment	1C-A	RECORD	Moisture (%) · Dry weight (kg)	WEIGHMENT_SLIP	Parallel stream.
LAB-PDY-S1	Paddy soak 1 — pit water	1C-A	RECORD	pH · EC (mS/cm) · TDS (ppm)	SAMPLE_PHOTO	Parallel stream. The water is tested, not the paddy.
LAB-PDY-S2	Paddy soak 2 — pit water	1C-A	RECORD	pH · EC (mS/cm) · TDS (ppm)	SAMPLE_PHOTO	Parallel stream.
LAB-PDY-S3	Paddy soak 3 — pit water	1C-A	RECORD	pH · EC (mS/cm) · TDS (ppm)	SAMPLE_PHOTO	Parallel stream.
LAB-PRE-T	Pre-Turner check	1C-B	RECORD	Moisture (%) · pH	SAMPLE_PHOTO	The last check before the material becomes piles.

Code	Checkpoint	Stage	Kind	Readings taken	Evidence	Note
LAB-T1-PRE	Moisture before T1	Turner	RECORD	Moisture (%)	SAMPLE_PHOTO	PER PILE. Six readings for a six-pile batch. Each tied to its own pile.
LAB-T1-POST	Moisture after T1	Turner	RECORD	Moisture (%)	SAMPLE_PHOTO	PER PILE. Six readings. This is the last Turner-stage laboratory work.
LAB-BNK-LOAD	Bunker loading	1C	GATE	Bunker height (m) · pH · EC (mS/cm) · Moisture (%) · Nitrogen (%) · Ash (%) · C:N ratio · Smell · Colour · Spring / squeeze test	BEFORE_PHOTO · HEIGHT_PHOTO	Height is measured FIRST — it sets the fill.
LAB-RLD-UNL	Bunker unloading — reload 1	1D-A	RECORD	Shrunken height (m) · Moisture (%)	HEIGHT_PHOTO	Shrunken height determines the reload height.
LAB-RLD-RE	Bunker reloading — reload 1	1D-A	RECORD	pH · EC (mS/cm) · Moisture (%) · Nitrogen (%) · Ash (%) · C:N ratio · Smell · Colour · Spring / squeeze test	SAMPLE_PHOTO	If moisture is short: water during handling, then re-measure pH, EC, moisture. The original reading is never overwritten.
LAB-RLD2-UNL	Bunker unloading — reload 2	1D-B	RECORD	Shrunken height (m) · Moisture (%)	HEIGHT_PHOTO	
LAB-RLD2-RE	Bunker reloading — reload 2	1D-B	RECORD	pH · EC (mS/cm) · Moisture (%) · Nitrogen (%) · Ash (%) · C:N ratio · Smell · Colour · Spring / squeeze test	SAMPLE_PHOTO	Only where the process runs a second reload.
LAB-TUN-PRE	Tunnel readiness	2A	RECORD	Tunnel height (m)	TUNNEL_PHOTO	Height checked first.
LAB-TUN-LOAD	Tunnel loading	2B	GATE	pH · EC (mS/cm) · Moisture (%) · Nitrogen (%) · Ash (%) · C:N ratio · Smell · Colour · Spring / squeeze test	BEFORE_PHOTO	Locked until height and the full set are approved.
LAB-QC-FINAL	Final QC — tunnel unloading	2D	RECORD	pH · EC (mS/cm) · Moisture (%) · Nitrogen (%) · Ash (%) · C:N ratio · Shrunken height (m) · Actinomycetes · Smell · Colour · Spring / squeeze test	SAMPLE_PHOTO · AFTER_PHOTO	The release record for the batch.

Reading the "Kind" column: 4 GATE · 1 DECISION · 20 RECORD. Reading the "Readings" column: 98 parameter readings per batch, before the two per-pile checkpoints are expanded across six piles.

7 • Checkpoint detail

One sheet per checkpoint, grouped by stage. Each sheet is what the technician is asked for, what proof is required with it, and what the reading can do.

Before the clock — raw material

LAB-RM-01 Raw material / bagasse weighment

RECORD · stage PRE-H0 · Incoming lot, before the clock starts

Readings	· Moisture (%) · pH · Dry weight (kg)
Evidence	LOT_PHOTO · WEIGHMENT_SLIP
Kind	RECORD — captured, never blocks
Note	ADVISORY. Recorded, never gates H0.

Stage 0A — wetting, and the first gate

LAB-WET-01 Bagasse wetting — first hopper pass

RECORD · stage 0A · During / at the end of hopper pass 1

Readings	· Moisture (%) · pH · EC (mS/cm)
Evidence	SAMPLE_PHOTO
Kind	RECORD — captured, never blocks
Note	Target moisture 69–71 %.

LAB-WET-02 Second hopper pass

RECORD · stage 0A · At the end of hopper pass 2

Readings	· Moisture (%) · pH · EC (mS/cm)
Evidence	SAMPLE_PHOTO
Kind	RECORD — captured, never blocks
Note	

LAB-BNK-PRE Before first bunker loading

GATE · stage 0A→0B · Immediately before bunker filling begins

Readings	· Moisture (%) · pH · EC (mS/cm)
Evidence	SAMPLE_PHOTO
Kind	GATE — the next activity is locked until an approved result exists
Note	Bunker filling stays locked until approved.

Stages 0B and 0C — bunker one and conditioning

LAB-UNL-01 Bunker unloading

RECORD · stage 0C · At unload, on the sample taken

Readings	· Moisture (%) · pH · EC (mS/cm)
Evidence	SAMPLE_PHOTO
Kind	RECORD — captured, never blocks
Note	

LAB-HOP-PRE Before hopper pass

RECORD · stage 0C · Between unload and the hopper pass

Readings	· Moisture (%) · pH · EC (mS/cm)
Evidence	SAMPLE_PHOTO
Kind	RECORD — captured, never blocks
Note	

LAB-RLD-01 Reloading to new bunker

RECORD · stage 0C · During reload

Readings	· Moisture (%) · pH · EC (mS/cm)
Evidence	SAMPLE_PHOTO
Kind	RECORD — captured, never blocks
Note	

LAB-MOIST-DEC Moisture decision point

DECISION · stage 0C→1B · After the second conditioning cycle

Readings	· Moisture (%)
Evidence	SAMPLE_PHOTO
Kind	DECISION — the reading chooses the branch
Note	≥ 68 % proceed. < 67 % controlled mist, then re-measure. 67–68 % is UNRESOLVED — record, flag, refer.

Stages 1A and 1B — chicken manure and the mix

LAB-CM-ARR Chicken manure — on arrival

RECORD · stage 1A · When the manure is received

Readings	• Moisture (%) • pH • Nitrogen (%) • Ash (%)
Evidence	LOT_PHOTO
Kind	RECORD — captured, never blocks
Note	<i>Initial composition of the incoming lot.</i>

LAB-CM-USE Chicken manure — before use in mixing

GATE · stage 1A→1B · On unload for the final mix

Readings	• Moisture (%) • pH • Nitrogen (%) • Ash (%)
Evidence	SAMPLE_PHOTO
Kind	GATE — the next activity is locked until an approved result exists
Note	<i>CONFIGURABLE. Required only when the gap since arrival exceeds the configured window. Where arrival is under 12 h before use, this checkpoint is skipped and the arrival reading stands.</i>

Stage 1C — paddy, the pre-Turner check, and bunker two

LAB-PDY-WGH Paddy weighment

RECORD · stage 1C-A · At bale weighment, before soaking

Readings	• Moisture (%) • Dry weight (kg)
Evidence	WEIGHMENT_SLIP
Kind	RECORD — captured, never blocks
Note	<i>Parallel stream.</i>

LAB-PDY-S1 Paddy soak 1 — pit water

RECORD · stage 1C-A · During soaking 1

Readings	• pH • EC (mS/cm) • TDS (ppm)
Evidence	SAMPLE_PHOTO
Kind	RECORD — captured, never blocks
Note	<i>Parallel stream. The water is tested, not the paddy.</i>

LAB-PDY-S2 Paddy soak 2 — pit water

RECORD · stage 1C-A · During soaking 2

Readings	• pH • EC (mS/cm) • TDS (ppm)
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Evidence	SAMPLE_PHOTO
Kind	RECORD — captured, never blocks
Note	Parallel stream.

LAB-PDY-S3 Paddy soak 3 — pit water

RECORD · stage 1C-A · During soaking 3

Readings	· pH · EC (mS/cm) · TDS (ppm)
Evidence	SAMPLE_PHOTO
Kind	RECORD — captured, never blocks
Note	Parallel stream.

LAB-PRE-T Pre-Turner check

RECORD · stage 1C-B · After the second yard flip, before the Turner begins

Readings	· Moisture (%) · pH
Evidence	SAMPLE_PHOTO
Kind	RECORD — captured, never blocks
Note	The last check before the material becomes piles.

LAB-BNK-LOAD Bunker loading

GATE · stage 1C · At each bunker fill — ONE PER BUNKER

Readings	· Bunker height (m) · pH · EC (mS/cm) · Moisture (%) · Nitrogen (%) · Ash (%) · C:N ratio · Smell — observation · Colour — observation · Spring / squeeze test — observation [Too dry Normal Too wet Dripping]
Evidence	BEFORE_PHOTO · HEIGHT_PHOTO
Kind	GATE — the next activity is locked until an approved result exists
Note	Height is measured FIRST — it sets the fill.

The Turner — before T1 and after T1, per pile

LAB-T1-PRE Moisture before T1

RECORD · stage Turner · After T0, before T1 — ONE READING PER PILE

Readings	· Moisture (%)
Evidence	SAMPLE_PHOTO

Kind	RECORD — captured, never blocks
Note	PER PILE. Six readings for a six-pile batch. Each tied to its own pile.

LAB-T1-POST Moisture after T1

RECORD · stage Turner · Immediately after T1 — ONE READING PER PILE

Readings	· Moisture (%)
Evidence	SAMPLE_PHOTO
Kind	RECORD — captured, never blocks
Note	PER PILE. Six readings. This is the last Turner-stage laboratory work.

Stage 1D — the two reloads

LAB-RLD-UNL Bunker unloading — reload 1

RECORD · stage 1D-A · At unload, before reloading

Readings	· Shrunk height (m) · Moisture (%)
Evidence	HEIGHT_PHOTO
Kind	RECORD — captured, never blocks
Note	Shrunk height determines the reload height.

LAB-RLD-RE Bunker reloading — reload 1

RECORD · stage 1D-A · During reload

Readings	· pH · EC (mS/cm) · Moisture (%) · Nitrogen (%) · Ash (%) · C:N ratio · Smell — observation · Colour — observation · Spring / squeeze test — observation [Too dry Normal Too wet Dripping]
Evidence	SAMPLE_PHOTO
Kind	RECORD — captured, never blocks
Note	If moisture is short: water during handling, then re-measure pH, EC, moisture. The original reading is never overwritten.

LAB-RLD2-UNL Bunker unloading — reload 2

RECORD · stage 1D-B · At unload, before the second reload

Readings	· Shrunk height (m) · Moisture (%)
Evidence	HEIGHT_PHOTO
Kind	RECORD — captured, never blocks
Note	

LAB-RLD2-RE Bunker reloading — reload 2

RECORD · stage 1D-B · During the second reload

Readings	· pH · EC (mS/cm) · Moisture (%) · Nitrogen (%) · Ash (%) · C:N ratio · Smell — observation · Colour — observation · Spring / squeeze test — observation [Too dry Normal Too wet Dripping]
Evidence	SAMPLE_PHOTO
Kind	RECORD — captured, never blocks
Note	Only where the process runs a second reload.

Stages 2A to 2D — tunnel, and the release record

LAB-TUN-PRE Tunnel readiness

RECORD · stage 2A · After tunnel preparation, before filling

Readings	· Tunnel height (m)
Evidence	TUNNEL_PHOTO
Kind	RECORD — captured, never blocks
Note	Height checked first.

LAB-TUN-LOAD Tunnel loading

GATE · stage 2B · During tunnel filling

Readings	· pH · EC (mS/cm) · Moisture (%) · Nitrogen (%) · Ash (%) · C:N ratio · Smell — observation · Colour — observation · Spring / squeeze test — observation [Too dry Normal Too wet Dripping]
Evidence	BEFORE_PHOTO
Kind	GATE — the next activity is locked until an approved result exists
Note	Locked until height and the full set are approved.

LAB-QC-FINAL Final QC — tunnel unloading

RECORD · stage 2D · At 24 °C, when unloading begins

Readings	· pH · EC (mS/cm) · Moisture (%) · Nitrogen (%) · Ash (%)
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	• C:N ratio • Shrunken height (m) • Actinomycetes — observation [Not visible Weak Clearly present] • Smell — observation • Colour — observation • Spring / squeeze test — observation [Too dry Normal Too wet Dripping]
Evidence	SAMPLE_PHOTO · AFTER_PHOTO
Kind	RECORD — captured, never blocks
Note	<i>The release record for the batch.</i>

8 • The parameter list

Every distinct reading the laboratory is asked for, and everywhere it appears. This is the list that must have a method behind it — section C to H of the appendix asks for that method, parameter by parameter.

Parameter	Unit	Type	Appears	At
Moisture	%	number	21×	RM-01 • WET-01 • WET-02 • BNK-PRE • UNL-01 • HOP-PRE • RLD-01 • CM-ARR • CM-USE • MOIST-DEC • PDY-WGH • PRE-T • T1-PRE • T1-POST • BNK-LOAD • RLD-UNL • RLD-RE • RLD2-UNL • RLD2-RE • TUN-LOAD • QC-FINAL
pH	—	number	18×	RM-01 • WET-01 • WET-02 • BNK-PRE • UNL-01 • HOP-PRE • RLD-01 • CM-ARR • CM-USE • PDY-S1 • PDY-S2 • PDY-S3 • PRE-T • BNK-LOAD • RLD-RE • RLD2-RE • TUN-LOAD • QC-FINAL
Dry weight	kg	number	2×	RM-01 • PDY-WGH
EC	mS/cm	number	14×	WET-01 • WET-02 • BNK-PRE • UNL-01 • HOP-PRE • RLD-01 • PDY-S1 • PDY-S2 • PDY-S3 • BNK-LOAD • RLD-RE • RLD2-RE • TUN-LOAD • QC-FINAL
Nitrogen	%	number	7×	CM-ARR • CM-USE • BNK-LOAD • RLD-RE • RLD2-RE • TUN-LOAD • QC-FINAL
Ash	%	number	7×	CM-ARR • CM-USE • BNK-LOAD • RLD-RE • RLD2-RE • TUN-LOAD • QC-FINAL
TDS	ppm	number	3×	PDY-S1 • PDY-S2 • PDY-S3
Bunker height	m	number	1×	BNK-LOAD
C:N ratio	—	number	5×	BNK-LOAD • RLD-RE • RLD2-RE • TUN-LOAD • QC-FINAL
Smell	—	observation	5×	BNK-LOAD • RLD-RE • RLD2-RE • TUN-LOAD • QC-FINAL
Colour	—	observation	5×	BNK-LOAD • RLD-RE • RLD2-RE • TUN-LOAD • QC-FINAL
Spring / squeeze test	—	observation	5×	BNK-LOAD • RLD-RE • RLD2-RE • TUN-LOAD • QC-FINAL
Shrunken height	m	number	3×	RLD-UNL • RLD2-UNL • QC-FINAL
Tunnel height	m	number	1×	TUN-PRE
Actinomycetes	—	observation	1×	QC-FINAL

What is missing from this table, deliberately

There is no method column and no threshold column, because the factory has not stated them.

A moisture percentage is meaningless until somebody says whether it is oven-dry at 105 °C for a stated time or a hand-held meter, and how many grams, and taken from where in the pile.

Two technicians using two methods produce two numbers that look identical in a database and are not comparable. That is the single most likely way this system produces confident nonsense.

The system holds a method field and a threshold band against every parameter, empty, waiting. Filling them is a laboratory exercise, not a build input — which is why the appendix does not ask for them. The appendix asks how long each test takes, because that is what the software has to schedule around.

9 • Evidence

A reading without a photograph is a claim. A reading with one is a record. Evidence is what makes a laboratory result defensible three months later, when nobody remembers the sample and the batch is already sold.

Evidence type	Required at	What it must show
LOT_PHOTO	2 checkpoints	The incoming lot as received, before it is disturbed.
WEIGHMENT_SLIP	2 checkpoints	The weighbridge or scale slip, readable.
SAMPLE_PHOTO	17 checkpoints	The physical sample with its label legible in frame.
BEFORE_PHOTO	2 checkpoints	—
HEIGHT_PHOTO	3 checkpoints	—
TUNNEL_PHOTO	1 checkpoints	—
AFTER_PHOTO	1 checkpoints	The material after the operation the reading describes.

Three rules about photographs

- The photograph is taken at the point of capture, on the device, at the time. A photograph selected from a gallery is not evidence of when — and the whole value of it is the when.
- The label must be legible in the frame. A photograph of compost is a photograph of compost; it identifies nothing.
- Evidence is never replaced. A better photograph is a new one that supersedes the old, and both are kept. Nothing in this system overwrites, and evidence least of all.

Status: AMBER, honestly

Evidence storage is implemented in the system but its twenty tests currently skip rather than pass, on a storage-key configuration problem.

"The code exists" is not the same as "it is proved". This document says AMBER and not GREEN because nothing has yet asserted, against a real database, that a client cannot delete an evidence object directly.

That proof is on the board and is not the laboratory's problem to solve — but the laboratory should know that today the photograph is trusted rather than protected.

10 • Entry, approval and what a gate actually waits for

The distinction that keeps this honest: entry and approval are two different acts, by two different people, at two different times. A gate waits for the second one.

Step	Who	What it means
CAPTURE	The technician	The reading is entered with its evidence, on the device, at the bench. It exists in the batch file from this moment and can never be removed from it.
APPROVE	UNRESOLVED — see Q M-1	Somebody states that the reading is fit to act on. Only this lifts a gate. Who may do it, and whether the technician may approve their own reading, is question M-1 and is not decided.
SUPERSEDE	The technician, with a reason	A retest replaces the operative value. The original stays visible, with the reason, in order. There is no edit and no delete anywhere in this system.

What a correction looks like

It looks like a second row. Not a changed first row.

#	Reading	Status	Why
1	Moisture 64.2 %	SUPERSEDED	Meter had not been calibrated that morning
2	Moisture 67.8 %	OPERATIVE	Retest on the retained sample after calibration

Both rows are in the batch file forever. Anyone can see that the number moved, by how much, and why. That is not bureaucracy — it is the only version of this that survives a customer complaint.

11 • Out of range, and what the system does about it

Thresholds are not written into this document, because the factory has not stated them. What is decided is the shape of the response, and that shape does not change when the numbers arrive.

IN BAND	Recorded. Nothing else happens.
OUT OF BAND	Recorded, flagged on the batch, and shown to the operator and the supervisor as a finished sentence — never as a raw comparison. The reading is not rejected and the technician is never asked to re-enter it until it looks acceptable.
IMPOSSIBLE	A moisture of 140 % is a typing error, not a finding. Range validation at entry catches the typo; it must be wide enough that a genuine surprise still gets through. A validator that refuses real readings teaches people to write down the number the machine will accept.

The failure mode this section exists to prevent

A system that scolds a technician for an out-of-range reading gets in-range readings. Always. Within about three weeks.

The reading is never the technician's fault, and the interface must never imply that it is. Out of range is information about the compost, not about the person holding the meter.

Question N-10 in the appendix asks the laboratory directly what happens to the person who reports a bad number. The answer to that one predicts the quality of every number this system will ever hold.

12 • Why the backend is frozen and this document is not

The laboratory will change its mind. Not once — repeatedly, and correctly, as it learns what the process actually does. A specification that treats that as a problem is a specification that will be abandoned by December.

The system is built so that those two facts do not fight. The engine is frozen. The laboratory specification is data.

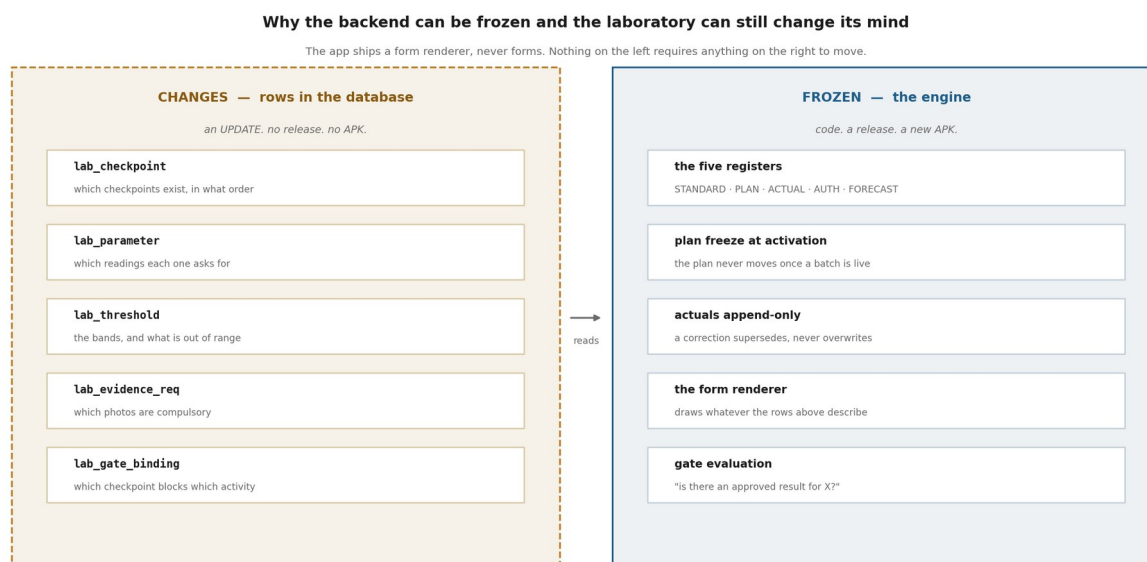


Figure 3 — what changes without a release, and what does not

What this means concretely

If the laboratory says...	Then...
"Add EC to the pre-Turner check"	One row. Live the same hour. No release, no APK, nobody reinstalls anything.
"Moisture at bunker loading is now a gate, not a record"	One row. Live the same hour.
"The chicken-manure window is 8 hours, not 12"	One value. Live the same hour.
"We now test after T2 as well"	Two rows — a checkpoint and its parameter. Live the same hour.
"Moisture must be oven-dry at 105 °C, and the meter reading is advisory"	A method field and a second parameter. Rows, not code.
"An extension must now be approved by two people, not one"	<i>That is the engine. A release. This is the boundary, and it is a narrow one.</i>

The one sentence that carries this

The app ships a form renderer, never forms.

It does not know what moisture is. It knows how to draw whatever the checkpoint rows describe, collect it, attach a photograph, and refuse to let it be edited afterwards.

That is why the operator and laboratory screens can ship tomorrow against a specification the laboratory has not finished writing.

How a change is actually made

Step		Detail
1	The laboratory says it	In writing, to the process owner. A verbal change is a change nobody can point at later.
2	It is recorded here	This document is versioned. LAB-2026A v1 becomes v2. The old version is kept, never edited.
3	The rows are updated	By an administrator, in the configuration screen. Effective from a stated date, never retroactively.
4	Live batches are unaffected	A batch that is already running keeps the specification it started under. This is the same rule as the plan freeze, and for the same reason.

Step 4 is the one people forget. If a checkpoint is added on day nine of a batch, that batch does not retroactively acquire a hole where the new checkpoint should have been. It runs under the specification it started with. The next batch gets the new one.

13 • Tomorrow — the minimum capture set

The apps go out tomorrow. A batch name is set, and the laboratory and the operators start capturing against it. What ships tomorrow is deliberately less than what this document describes.

Tomorrow, nothing gates.

The four gates are described in this document and built into the data — and they are switched off for the first batch.

A gate that fires on day one, against thresholds nobody has confirmed, against a method nobody has written down, stops the factory for a reason that turns out to be wrong. That happens once and the app is never opened again.

The first batch is for learning what the readings actually look like. Gates come on when the laboratory has signed the thresholds — section 11 and section 14.

What the app must capture tomorrow

Batch	The batch name, exactly as the factory says it out loud. Chosen once, on the day, and used by everybody.
Checkpoint	Which of the 25 this is. Chosen from a list, never typed.
Pile	For the two per-pile checkpoints only. 1 to 6.
Values	Whatever the checkpoint asks for. Numbers as numbers. Observations from a list where the list exists, free text where it does not.
Photograph	Taken on the device, at the bench, with the label legible.
Who	From the login. Never typed, never chosen from a dropdown of names.
When	The device clock at capture, and the server clock at receipt. Both. They will disagree and the disagreement is worth knowing about.

What is explicitly not in tomorrow's app

- Thresholds and out-of-range colouring — no confirmed numbers to colour against.
- Gate enforcement — described above.
- Approval as a separate act — capture only. Approval arrives when M-1 is answered.
- The trend screen — twenty readings is not a trend. It arrives when there is something to show.

This list is not a compromise. Capturing six honest fields against a real batch, from tomorrow, is worth more than a complete application shipped in three weeks — because in three weeks the questions in the appendix will still be unanswered, and the only thing that answers them is watching real readings arrive.

14 • What this document does not decide

Six things are open. They are listed here rather than resolved with a sensible default, because a sensible default becomes invisible within a week and gets defended as a factory decision within a month.

Open item	Where	Why it is left open	Question
The 67-68 % moisture band	LAB-MOIST-DEC	The process states ≥ 68 proceed and < 67 mist. The band between them has no rule. No gate is built on it and the system refers it to a human.	L-2
Who approves a laboratory result	all four gates	Whether the technician may approve their own reading, and who else may. Until answered, tomorrow's app captures only and nothing is gated.	L-5
Every threshold, for every parameter	§8, §11	No band in this document is confirmed. Out-of-range behaviour is defined; the ranges are not. The table holds them; the laboratory has to state them.	not asked here
The method behind each reading	§8	Oven versus meter, sample mass, drying time, calibration interval. Two methods produce two incomparable numbers that look the same in a database. A laboratory exercise, not a build input.	not asked here
Whether a gate can be overridden	§2	And by whom, and whether it needs a reason. The system implements no override until this is answered.	M-4
The chicken-manure window measurement point	LAB-CM-USE	Arrival-to-unload or arrival-to-mixing-start. The 12-hour value is configurable; what it is measured between is not yet stated.	J-2

The standing rule

Nothing in this project may invent a factory rule. If a build needs one of the six above, it stops and asks.

A number that arrived by inference, wearing the clothes of a decision, is the most expensive artefact this system can produce.

15 • Sign-off

By signing, each party states that the sections named against their role describe what the laboratory and the factory actually do, or intend to do, as of the date of signature.

A signature here is not a statement that the thresholds are correct — section 14 records that they do not yet exist. It is a statement that the shape is right: these checkpoints, these readings, these four gates, this Turner rule, this change process.

Role	Name	Signature	Date
Laboratory in charge	sections 1–11, 14		
Laboratory technician	sections 2, 7, 9, 13		
Production manager	sections 2, 3, 13		
Process owner	all sections		
Quality / compliance	sections 9, 10, 12		
General manager	sections 12, 14		

Version

Document	LAB-2026A — The Laboratory Process, version 1
Companion	PROCESS-2026C — the 470-hour standard process (governs where the two disagree)
Generated from	<i>lab_checkpoints.json · questions.json — no figure in this document was typed by hand</i>
Supersedes	lab_technician_batch_process.md (working note, 30 Aug 2026)
Next review	On the first signed threshold, or after batch 1, whichever comes first

Appendix A • The timing worksheet

One row per checkpoint, and four numbers per row. These four numbers are what the software actually schedules with — everything else in this document tells it what to draw, and this tells it when things can happen.

Duration	Sample in hand to a number that exists. Include the waiting, not just the working. If it is a range, write the range — the system has to plan for the top of it.
People	How many are needed at the same time. One who then leaves it running is different from one who must stand there.
Production waits?	Does anybody stand still while this happens, or does work carry on and the number catch up later? This is the column that decides whether a checkpoint is worth gating.
Notice needed	How long before the sample is due does the laboratory need to be told. This is the one number that costs nothing to give and saves the most time.

Rough is fine. "About 40 min", "half a day", "we have never timed it" are all real answers and all three are more useful than a blank.

Code	Checkpoint	Stage	Kind	Duration (sample → number)	People	Prod. waits?	Notice needed	Notes
LAB-RM-01	Raw material / bagasse weighment	PRE -H0	RECORD					
LAB-WET-01	Bagasse wetting — first hopper pass	0A	RECORD					
LAB-WET-02	Second hopper pass	0A	RECORD					
LAB-BNK-PRE	Before first bunker loading	0A→0B	GATE					
LAB-UNL-01	Bunker unloading	0C	RECORD					
LAB-HOP-PRE	Before hopper pass	0C	RECORD					
LAB-RLD-01	Reloading to new bunker	0C	RECORD					
LAB-CM-ARR	Chicken manure — on arrival	1A	RECORD					
LAB-CM-USE	Chicken manure — before use in mixing	1A→1B	GATE					
LAB-MOIST-DEC	Moisture decision point	0C→1B	DECISION					

Code	Checkpoint	Stage	Kind	Duration (sample → number)	People	Prod. waits?	Notice needed	Notes
LAB-PDY-WGH	Paddy weighment	1C-A	RECORD					
LAB-PDY-S1	Paddy soak 1 — pit water	1C-A	RECORD					
LAB-PDY-S2	Paddy soak 2 — pit water	1C-A	RECORD					
LAB-PDY-S3	Paddy soak 3 — pit water	1C-A	RECORD					
LAB-PRE-T	Pre-Turner check	1C-B	RECORD					
LAB-T1-PRE	Moisture before T1	Turner	RECORD					
LAB-T1-POST	Moisture after T1	Turner	RECORD					
LAB-BNK-LOAD	Bunker loading	1C	GATE					
LAB-RLD-UNL	Bunker unloading — reload 1	1D-A	RECORD					
LAB-RLD-RE	Bunker reloading — reload 1	1D-A	RECORD					
LAB-RLD2-UNL	Bunker unloading — reload 2	1D-B	RECORD					
LAB-RLD2-RE	Bunker reloading — reload 2	1D-B	RECORD					
LAB-TUN-PRE	Tunnel readiness	2A	RECORD					
LAB-TUN-LOAD	Tunnel loading	2B	GATE					
LAB-QC-FINAL	Final QC — tunnel unloading	2D	RECORD					

Two rows deserve a second line: LAB-T1-PRE and LAB-T1-POST are taken once per pile. Write the time for one pile and the time for all six — they are not six times apart, and the difference matters.

Appendix B • 189 questions about what the laboratory does

This appendix is the real deliverable. Everything before it is what the laboratory measures. This is how it actually works — what happens, in what order, how long each thing takes, who does it, what waits on what, and what it costs in hours when something goes wrong.

What these questions are for, and what they deliberately avoid

The software does not need to know how the laboratory labels a jar, or whether the technician writes in pen.

It needs durations. It needs to know that moisture takes forty minutes and nitrogen takes four hours, because that is what decides whether a gate stalls the line or nobody notices it.

So none of these ask about bench convention. Every one of them answers: what do you do · in what order · how long · who · what waits on what · how many at once · and what does it cost in time when it goes wrong.

Tag	Count	Meaning
TONIGHT	80	Shapes what gets built. A rough answer beats a precise one that arrives next week — "about forty minutes" is a usable number and "we have never timed it" is a usable answer too.
WEEK 1	89	Needed before the first batch finishes. Not blocking the build; blocking the schedule being right.
LATER	20	Shapes the system over the next quarter. Ask while the answers are cheap.

How to use this appendix

Print it. Stand in the laboratory for one shift. Write on it.

Where the answer is a range, write the range — "twenty minutes to two hours" tells the system more than an average does, because the system has to plan for the two hours.

Where the answer is "we have never timed it", write that. It is a real answer, and it means somebody should time it once during the first batch.

Where the answer is "we don't do that", write that too. An absence recorded is worth as much as a rule recorded — section 3 of this document exists entirely because somebody said "there is no test after T0".

A • One batch, end to end

Before any screen is drawn, somebody has to be able to describe a batch from the laboratory's side as a sequence of things that happen and take time. This section is that description.

A-1	TONIGHT	Walk one batch from start to finish from the laboratory's side. What is the first thing you do, and the last?
A-2	TONIGHT	How many times does the laboratory get involved in one batch? Roughly, not exactly.
A-3	TONIGHT	Across the whole batch, how many hours of laboratory work is that in total?
A-4	TONIGHT	Which of those involvements is the longest, and how long?
A-5	TONIGHT	Which one is the most likely to make production wait?
A-6	WEEK 1	Is the pattern the same for every batch, or does it change with the season or the raw material?
A-7	WEEK 1	How many days of the 20 does the laboratory touch the batch at all? Which days are quiet?

A-8	WEEK 1	Is there a day where the laboratory is doing several things for the same batch at once?
A-9	WEEK 1	What does a normal day look like when two batches are running?
A-10	WEEK 1	Is any part of the batch handled by someone other than the laboratory — production taking its own reading, for example?
A-11	LATER	What does the laboratory do for a batch that nobody has asked for, but that is done anyway?
A-12	LATER	What part of the batch does the laboratory feel it has no visibility of?

B • How long each test takes

The single most useful set of numbers in this whole document. Every scheduling decision the system makes depends on knowing how long a test takes — and "a test" means from sample in hand to a number that exists.

B-1	TONIGHT	Moisture: how long from sample in hand to a number, start to finish?
B-2	TONIGHT	pH: how long?
B-3	TONIGHT	EC: how long?
B-4	TONIGHT	TDS: how long?
B-5	TONIGHT	Nitrogen: how long?
B-6	TONIGHT	Ash: how long?
B-7	TONIGHT	C:N — is it calculated from the above, or is it its own test with its own time?
B-8	TONIGHT	Dry weight: how long?
B-9	TONIGHT	The visual checks — smell, colour, squeeze, actinomycetes: how long, together?
B-10	TONIGHT	Heights — bunker, tunnel, shrunken: how long, and is it the laboratory that measures them or production?
B-11	WEEK 1	For each of the above: how much of that time is the machine working, and how much is a person working?
B-12	WEEK 1	Which of these can be left running unattended while something else is done?
B-13	WEEK 1	Which of these needs the technician to stand there for the whole time?
B-14	WEEK 1	If all the tests at one checkpoint are done together, is the total the sum of the parts, or less?
B-15	WEEK 1	Is the time different for the first sample of the day versus the fifth?
B-16	LATER	Which test would you most like to be faster, and what makes it slow?

C • From sample to number — the gap that decides everything

The system needs to know not how long a test takes, but how long production waits. Those are different numbers and the difference is where the whole schedule lives.

C-1	TONIGHT	For each checkpoint: how long between the sample being taken and production being told the answer?
C-2	TONIGHT	Is that gap minutes, hours, or the next morning? Give the honest normal, not the best case.
C-3	TONIGHT	Which checkpoints have a gap long enough that production has already moved on by the time the number exists?
C-4	TONIGHT	Is production ever waiting, physically standing there, for a laboratory number? Which checkpoints?

C-5	WEEK 1	When production is waiting, what is the longest it has actually waited?
C-6	WEEK 1	Is the sample taken as soon as the material is ready, or at a fixed time?
C-7	WEEK 1	How long can a sample sit before it is tested without the number changing?
C-8	WEEK 1	Does the answer to that differ by test? Moisture especially.
C-9	WEEK 1	Is the number ever given verbally before it is written down anywhere?
C-10	WEEK 1	How much earlier could the laboratory be told a sample is coming, and would that help?
C-11	LATER	If the laboratory knew the whole day's sampling schedule at 6am, what would change?
C-12	LATER	Which gap, if halved, would make the biggest difference to production?

D • Sequence and dependencies — what waits on what

This is the section that decides whether the system can schedule laboratory work at all, or only record it after the fact.

D-1	TONIGHT	Which laboratory results must exist before production can move to the next step? Name them.
D-2	TONIGHT	For each of those, what physically happens if the result is not there yet — does the material sit, or does work continue?
D-3	TONIGHT	Is any laboratory test dependent on another test finishing first?
D-4	WEEK 1	Does the laboratory ever hold a sample until a related sample arrives so both can be run together?
D-5	WEEK 1	Is there a test that only makes sense if an earlier one came back a certain way?
D-6	WEEK 1	Does an out-of-range result change what is tested next, or only what production does next?
D-7	WEEK 1	Which laboratory work is time-critical — must happen within a window — and which can slide by a day?
D-8	WEEK 1	For the ones with a window, how wide is the window?
D-9	WEEK 1	What has to be prepared in advance — reagents, calibration, drying — and how long before?
D-10	LATER	Is anything tested that nothing downstream actually uses?

E • What runs at the same time

The process has real parallelism — paddy soaking alongside the main rail, six piles under two machines. The laboratory's parallelism is what decides whether that is a problem.

E-1	TONIGHT	Can the laboratory run more than one test at the same time? Which combinations?
E-2	TONIGHT	Can two different batches be tested in the same session?
E-3	TONIGHT	When six piles need the same test, are they done one after another or all together?
E-4	WEEK 1	How much longer does six piles take than one pile? Six times, or less?
E-5	WEEK 1	Is the paddy stream tested by the same person at the same bench as the main line?
E-6	WEEK 1	Does testing the paddy soak ever collide with testing the main line? What gives way?

E-7	WEEK 1	If two checkpoints fall due at the same hour, which one is done first, and who decides?
E-8	WEEK 1	Is there a queue today, and how long does it get?
E-9	LATER	What is the most the laboratory has ever had waiting at once?

F • Equipment, and how many at once

Capacity is a hard ceiling the system cannot schedule around unless it knows where it is.

F-1	TONIGHT	How many samples can the oven hold at once?
F-2	TONIGHT	How many moisture meters are there, and how many people can use them at the same time?
F-3	TONIGHT	How many pH and EC meters?
F-4	TONIGHT	Is there one balance, or several?
F-5	WEEK 1	Which piece of equipment is the bottleneck on a busy day?
F-6	WEEK 1	How long does the oven take to be free again after a run?
F-7	WEEK 1	Does anything need to cool, settle or stabilise before the next use, and for how long?
F-8	WEEK 1	What happens when a meter is being calibrated — does testing stop?
F-9	WEEK 1	How often does calibration happen, and how long does it take?
F-10	WEEK 1	Is any test sent outside the factory? Which, and what is the turnaround in days?
F-11	LATER	If one more of any instrument were bought, which one, and how much time would it save?

G • People, shifts and cover

The system will ask somebody to do something at H317, which may be four in the morning. It needs to know whether anybody is there.

G-1	TONIGHT	How many people work in the laboratory?
G-2	TONIGHT	What hours does the laboratory actually work? Start and end.
G-3	TONIGHT	Is there anybody in the laboratory at night?
G-4	TONIGHT	What happens to a checkpoint that falls due at 2am?
G-5	TONIGHT	Sundays and holidays — does the laboratory run?
G-6	WEEK 1	Can any of these tests be done by a production operator rather than a technician? Which?
G-7	WEEK 1	Who covers when the technician is on leave or sick?
G-8	WEEK 1	Is there a handover between shifts, and what is passed on?
G-9	WEEK 1	Does the same person do the sampling and the testing, or two people?
G-10	WEEK 1	How long does it take a new person to be trusted with each test?
G-11	LATER	Is there work the laboratory does that is not testing at all — cleaning, preparation, paperwork — and how many hours a day?

H • The checkpoints, one at a time

For each of the 25, the system needs the same four numbers. Answer them as a row per checkpoint rather than as prose — a table is fine, a scribbled list is fine.

H-1	TONIGHT	For each checkpoint: how long does the whole thing take, sample to number?
H-2	TONIGHT	For each checkpoint: how many people are needed?
H-3	TONIGHT	For each checkpoint: does production wait, or carry on?
H-4	TONIGHT	For each checkpoint: how much warning does the laboratory need that it is coming?
H-5	WEEK 1	Which of the 25 are done at a fixed time of day regardless of the batch?
H-6	WEEK 1	Which of the 25 are done "when production calls"?
H-7	WEEK 1	Are any of the 25 done together as one visit to the floor?
H-8	WEEK 1	Are any of the 25 skipped in practice, and under what circumstances?
H-9	WEEK 1	Are any of the 25 done more often than the process says?
H-10	WEEK 1	Is there a checkpoint the laboratory does that is not in the list of 25?
H-11	LATER	Which of the 25 does the laboratory consider least useful?

I • The Turner window

Two readings, six piles, two machines, and a process that does not stop. This is the tightest timing in the whole batch and the one most likely to be got wrong in software.

I-1	TONIGHT	Before T1 and after T1, six piles each — how long does the whole set of six take?
I-2	TONIGHT	Is the "before T1" set taken all at once, or pile by pile as each pile is about to be turned?
I-3	TONIGHT	How long before the T1 pass does the "before" sample need to be taken?
I-4	TONIGHT	How soon after a pile is turned is the "after" sample taken?
I-5	TONIGHT	Does the turner wait for the laboratory at any point, or never?
I-6	WEEK 1	If the laboratory is not ready, does T1 go ahead without the reading?
I-7	WEEK 1	Two machines turning at once — does that double the sampling load in the same hour?
I-8	WEEK 1	Is the sample taken from the pile before the machine reaches it, or after it has passed?
I-9	WEEK 1	Is a pile ever sampled twice because the first sample was taken at the wrong moment?
I-10	WEEK 1	How long does the whole T1 pass take across six piles, from the laboratory's point of view?
I-11	LATER	Would the laboratory want a reading at T0, T2 or T3 if it were free? What would it tell you?

J • Chicken manure and the 12-hour window

The one rule in the process that is already known to be conditional. The system needs to know exactly what the condition is measured between.

J-1	TONIGHT	Under 12 hours between arrival and use means no re-test. Is 12 the right number?
J-2	TONIGHT	Measured from arrival to what, exactly — unloading, mixing start, or something else?
J-3	TONIGHT	How long does the arrival test itself take, from lorry to number?
J-4	TONIGHT	Does the lorry wait for the arrival test, or is it unloaded first?

J-5	WEEK 1	If the manure sits longer than the window, is the full set repeated or only part of it?
J-6	WEEK 1	How often does manure actually sit longer than the window? Most loads, or rarely?
J-7	WEEK 1	Does manure arrive in one load or several, and does each load get its own test?
J-8	WEEK 1	How much notice does the laboratory get that manure is arriving?
J-9	WEEK 1	Does covered versus uncovered storage change how fast it changes?
J-10	LATER	Has a load ever been rejected on arrival, and what happened to the batch that day?

K • When something goes wrong, and what it costs in time

Rework is a duration, and the system has to be able to plan for it. Every answer here should be a number of hours.

K-1	TONIGHT	A result comes back out of range. What happens next, and how long does it take?
K-2	TONIGHT	A retest — how long from deciding to retest to having the new number?
K-3	TONIGHT	Moisture is low and misting is needed. How long is the misting, and how long before you re-measure?
K-4	TONIGHT	After misting, is the re-measure the full test again, or a quicker check?
K-5	WEEK 1	A sample is lost or spoiled. How long to get another one?
K-6	WEEK 1	An instrument fails mid-batch. What happens, and how long is the delay?
K-7	WEEK 1	Power fails during a drying run — is the result void, and how much time is lost?
K-8	WEEK 1	How often does a checkpoint have to be repeated? Once a batch, once a month?
K-9	WEEK 1	A batch is held longer than planned. Do any checkpoints have to be repeated, and which?
K-10	WEEK 1	How long does the laboratory keep a sample in case it is needed again?
K-11	LATER	What is the longest a laboratory problem has ever delayed a batch?

L • Decisions the laboratory makes

Some readings are recorded. Some readings change what happens next. The system must know which is which, who decides, and how long the deciding takes.

L-1	TONIGHT	Which laboratory readings actually change what production does next? Name them.
L-2	TONIGHT	The moisture decision: at or above 68 proceed, below 67 mist. What happens between 67 and 68?
L-3	TONIGHT	Who makes that call — the technician, the laboratory in charge, or production?
L-4	TONIGHT	How long does that decision normally take once the number exists?
L-5	TONIGHT	Does anyone have to approve a laboratory result before production acts on it?
L-6	WEEK 1	If yes, who, and how long does that normally take?
L-7	WEEK 1	At night and at weekends, who makes the call?
L-8	WEEK 1	Is a decision ever made on a number before it is formally recorded?
L-9	WEEK 1	Does production ever overrule the laboratory? What happens then?
L-10	WEEK 1	How is the decision communicated — phone, in person, a book?

L-11	LATER	Is there a decision the laboratory would like to make but currently cannot?
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M • What stops production, and for how long

Four checkpoints in the standard are marked as gates. This section checks whether they really are, and what happens on the floor when one bites.

M-1	TONIGHT	Which laboratory results, if missing, actually stop production today?
M-2	TONIGHT	When that happens, what does production do — wait, or continue and record it?
M-3	TONIGHT	How long has production actually been stopped by the laboratory, at most?
M-4	TONIGHT	Can anybody let production continue without the result? Who?
M-5	WEEK 1	When that override happens, is anything written down?
M-6	WEEK 1	Are there results the laboratory thinks should stop production but currently do not?
M-7	WEEK 1	Are any of the four gates not really gates in practice?
M-8	WEEK 1	Would a warning an hour before a gate is due be useful, or noise?
M-9	LATER	Should a result ever be too old to rely on? After how long?

N • How the number travels today

Understanding the current path from bench to decision is what tells us what the app has to replace, and what it must not break.

N-1	TONIGHT	When a number exists, how does the person who needs it find out?
N-2	TONIGHT	How long does that take — immediately, at the end of the shift, next morning?
N-3	TONIGHT	Where is the number written down first?
N-4	TONIGHT	Does the same number get written more than once, in more than one place?
N-5	WEEK 1	Who else sees it, and when?
N-6	WEEK 1	Is anything typed into a computer today? What, and by whom, and when?
N-7	WEEK 1	At the end of a batch, is there a summary of all its laboratory results anywhere?
N-8	WEEK 1	How long does producing that summary take?
N-9	WEEK 1	Has a number ever gone missing between the bench and the person who needed it?
N-10	WEEK 1	When an out-of-range number is reported, what happens to the person who reported it — nothing, a question, or a problem? Answer this one plainly; it predicts the quality of every number the system will collect.
N-11	LATER	What does management ask the laboratory for, and how long does answering take?

O • What makes a test take longer than usual

Averages are useless for scheduling if the variance is large. This section is about the spread, not the middle.

O-1	TONIGHT	On a bad day, how much longer does a moisture result take than on a good day?
O-2	WEEK 1	What is the usual cause of a delay — waiting for the sample, waiting for equipment, or waiting for a person?
O-3	WEEK 1	Does the time of day change how long things take?
O-4	WEEK 1	Does the weather change anything — humidity, rain, the material being wetter?

0-5	WEEK 1	Is the first batch of a season slower than a settled one?
0-6	WEEK 1	Does a new technician change these times, and by how much?
0-7	WEEK 1	Which test has the most unpredictable duration?
0-8	LATER	If you had to promise production a time for each result, what would you promise?

P • Capacity — where the laboratory breaks

The factory intends to run more batches. The system should be able to say, in advance, when that stops being possible.

P-1	TONIGHT	How many batches can run at once before the laboratory cannot keep up?
P-2	TONIGHT	On the busiest day of a batch, how many hours of laboratory work is there?
P-3	WEEK 1	Which day of the 20 is the busiest for the laboratory, and why?
P-4	WEEK 1	If two batches were offset by a few days, would that help or make it worse?
P-5	WEEK 1	What is the first thing that would be dropped if the laboratory ran out of time?
P-6	WEEK 1	Is any of the current work done outside normal hours because there is no room in the day?
P-7	LATER	At what number of simultaneous batches does the laboratory need another person?
P-8	LATER	At what number does it need another instrument?

Q • What the laboratory is waiting on from everybody else

The laboratory is not only a source of delay; it is usually also a victim of one. The system can fix some of these for free.

Q-1	TONIGHT	What does the laboratory most often wait for that is not its own doing?
Q-2	TONIGHT	How much notice does the laboratory get before a sample is needed today?
Q-3	TONIGHT	How much notice would be enough?
Q-4	WEEK 1	Is the laboratory ever told about a batch after it has already started?
Q-5	WEEK 1	Does the laboratory know the plan for the next 24 hours? How?
Q-6	WEEK 1	When production changes the plan, how does the laboratory find out?
Q-7	WEEK 1	Is the laboratory ever asked for a test that was not planned, and how often?
Q-8	LATER	What one piece of information, given earlier, would save the laboratory the most time?

R • Tomorrow — what the app has to do on day one

These are the only questions that must be answered before the APK is built. Everything above can be answered during the first batch; these cannot.

R-1	TONIGHT	What is the batch identifier for tomorrow's batch? The exact text, as it will be said out loud.
R-2	TONIGHT	What stage will that batch be at when the app is handed over?
R-3	TONIGHT	Which checkpoints will realistically be hit in the first 48 hours?
R-4	TONIGHT	How many phones, and who gets which — laboratory, operator, supervisor?
R-5	TONIGHT	Who trains them, and how long does that take?
R-6	TONIGHT	Is there signal where the samples are taken and where the tests are done?
R-7	TONIGHT	Will they also keep the paper record tomorrow, in parallel? (They should.)

R-8	TONIGHT	If the app is wrong or confusing tomorrow, who do they call?
R-9	TONIGHT	If the app fails entirely, does work stop or continue on paper?
R-10	TONIGHT	What single thing, at the end of tomorrow, would make you call it a success?

Total: 189 questions · 80 tonight · 89 week one · 20 later — *generated from questions.json, not typed.*